CSA5111 - CRYPTOGRAPHY AND NETWORK SECURITY

CO-1 AT-II - CONCEPT MAPPING

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1. Security Attacks, Threats, Risks and Security Goals

Security Attacks (Actions by an attacker)
* Malware/virus
* Phishing, DDoS
* SQL Injection

leads to

Threats (Potential cause of harm)
* Data theft
* Unauthorized access
* Service disruption
* Data tampering

result in

Risks (Actual / Possible impact)
* Loss of Confidentiality
* Loss of integrity
* Loss of availability
* Financial loss

help mitigate

Security Goals (Mitigate the threats and risks)

- **Confidentiality**: Prevents unauthorized access to data
- **Integrity**: Ensures data is not altered illegitimately
- **Availability**: Ensures system & data are accesible

Real world Examples

Phishing Attack:
* Leads to Threat: Unauthorised access → Risk: Data theft
* Ex: Fake email steals user credentials

SQL Injection
* Leads to Threat: Data tampering → Risk: Integrity Loss
* Ex: Attacker injects SQL to modify DB records

DDoS Attack
* Leads to Threat: Service disruption → Risk: Availability loss
* Ex: Flooding server with traffic goes down

Security Goals Mitigation:
Ex: Encryption, I/p validation, Redundancy & backups

2. Comparison of Substitution Techniques

Substitution Techniques

Caesar cipher
Key Usage:
* Single numeric key (shift value 0-25)

Complexity:
* Very low

Security Strength
* Weak

Monoalphabetic Cipher
Key Usage:
* 26 letter permutation (one key)

Complexity:
* low

Security Strength
* Weak

Vigenere Cipher
Key Usage:
* Repeating keywords

Complexity:
* Medium

Security Strength:
* Moderate

Hill Cipher
Key Usage:
* Matrix key

Complexity:
* High

Security Strength
* Strong

Most Secure Technique: Hill Cipher
Why? It uses matrix operations and combines multiple characters, making it resistant to simple frequency analysis & statistical attacks, unlike other techniques

3. Classical Encryption schemes - Substitution vs Transposition

Classical Encryption Schemes

- Substitution Techniques
  - Caesar Cipher
  - Monoalphabetic cipher
  - Vigenere Cipher
  - Hill Cipher
- Transposition Technique
  - Rail fence Cipher
  - Columnar Transposition
  - Playfair
  - Route cipher

Difference	Similarities
* Substitution: changes the actual characters * Transposition: changes the order of characters * Substitution affects character identity * Transposition keeps characters but rearranges them	* Both are classical techniques * Both use key for encryption/decryption * Both are vulnerable to frequency analysis

Effects on Data Structure & Security Level

Technique	Data Structure	Security Level
Substitution	Same len, character changed	Lower
Transposition	Same characters reordered	Slightly higher

4. Steganography Concepts

Steganography

- **Data Hiding** (forms a type of Steganography)
  * Hides secret msg in a cover file
  * Maintains cover apperance
  * Goal: Convert Communication
- **Watermarking** (enhances Steganography)
  * Embeds visible/invisible mark
  * Used for authenticity & ownership
  * Goal: Protect content ownership
- **Survivability** (enhances Steganography)
  * Ensures hidden data/mark survives attacks
  * Uses robust embedding
  * Goal: Long-term reliability

How they Contribute to Secure Communication

Data Hiding
* Keeps Communication Secret
* Avoids detection by attackers

Watermarking
* Proves authenticity & ownership
* Prevents illegal copying
* Digital rights management (DRM)

Survivability
* Ensures data remains intact
* Even after modifications.